

AXYS processed dataset blueprint

Female-only gait kinematics joined to victimization survey labels.

- Working file: data/processed/Data_structure_all_subs.mat (83.3 MB, MATLAB v7)
- Originals: data/raw/
- Date compiled: 13 August 2026

1. Snapshot

Metric	Value
Subjects	31 females
Label split	17 victimized (Y) / 14 not (N)
Walking trials (WU*)	260
Kinematic sample rate	100 Hz
Force-plate rate (recorded, traces not stored)	1000 Hz
Trial duration	~1.17–4.60 s (117–460 frames)
Foot contacts per trial	2–8
Mass	47.7–100 kg (mean 64.1)
Height	141–168 cm (mean 160.0)

What changed from raw

- 12 male subjects removed.
- Survey columns from `Victimization surveys.xlsx` attached to each remaining subject.
- Join key: Excel `Subject No` = MATLAB `S{n}` (example: 14 → `Dat.S14`).
- Cohort-level `Dat.Res` dropped (it mixed sexes). Per-subject gait `Res` remains.

2. Files

Role	Path	Contents
Raw MAT	data/raw/Data_structure_all_subs.mat	43 subjects (31 F + 12 M), 111 MB
Raw survey	data/raw/Victimization surveys.xlsx	Subjects 1–50; sex, victimization, type, times, cyber
Processed MAT	data/processed/Data_structure_all_subs.mat	31 females + Survey table; labels on each <code>Sn</code>

Join rule

Use Excel column **Subject No**, not column **No**.

- Subject `No` is the lab ID and matches `Dat.S{n}`.
- `No` is only a 1–31 female roster index (`Survey.RosterNo`).

3. Survey field map

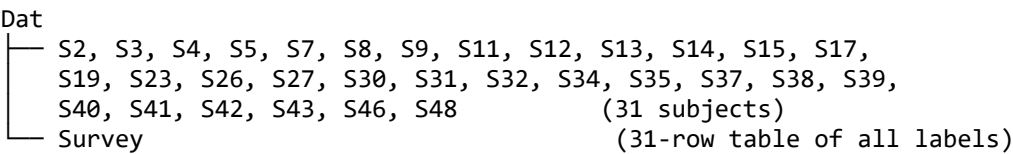
Excel column	MATLAB field	Meaning
No	Survey.RosterNo	Female roster 1–31
Subject No	Survey.SubjectNo	Lab ID; matches Sn
SEX	Survey.Sex	Always F in processed file
VICTIMIZED	Survey.Victimized	Y or N — primary ML label
person/online/both/ND/NO	Survey.VictimType	Ip, online, Both, Nd, or No
How many times	Survey.Times	Count or Nd; 0 if not victimized
CYBER BULLIED	Survey.CyberBullied	Yes, No, or Nd

Victim type among Y = 17: Nd 7, in-person (Ip) 6, Both 3, online 1.

CyberBullied overall: Yes 6, Nd 6, No 19 (No includes all 14 non-victims).

4. Top-level MATLAB layout

One variable: Dat.

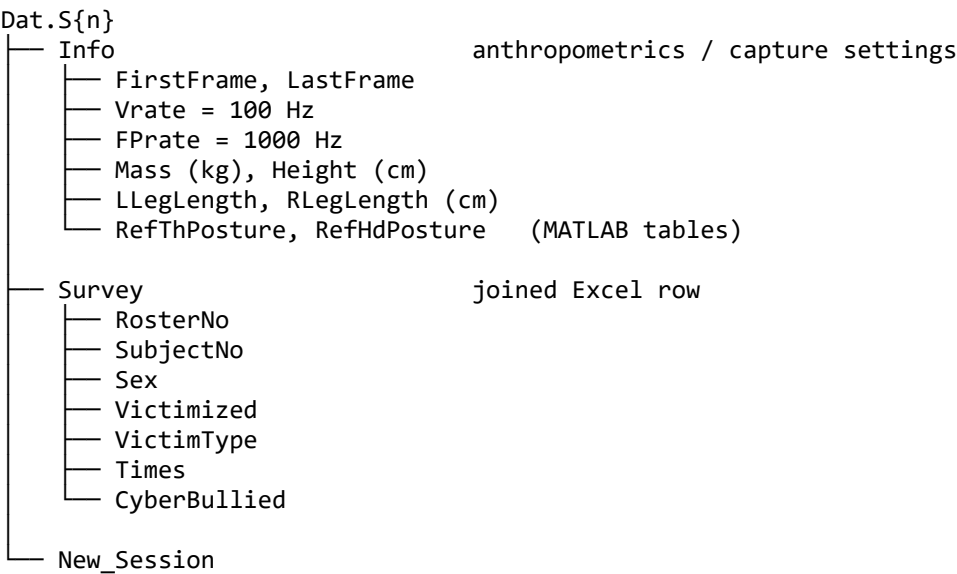


IDs not in the processed file: 1, 6, 10, 16, 18, 20–22, 24, 25, 28, 29, 33, 36, 44, 45, 47, 49, 50 (males, empty survey rows, or “DO NOT USE”).

Dat.Survey is a MATLAB table (opens in MATLAB). SciPy cannot decode that table or MATLAB string objects.

5. Per-subject tree

Every subject has three branches: Info, Survey, New_Session.



WU*	walking trials (count varies, typically 4-12)
kinematics	86 signals, each (n_frames × 3)
Info	gait events
KinFC	foot contact [frame, side]
KinFO	foot off
Midsvnt	mid-stance
Res	trial gait metrics (MATLAB tables)
static	quiet-standing calibration
Res	subject-level summary metrics
RawRes	unaggregated trial metrics

Event arrays (KinFC, KinFO, Midsvnt): size k × 2. Column 1 = frame index. Column 2 = side (1 or 2, left/right).
Example S2 WU01 KinFC:

```
15 1
71 2
127 1
183 2
236 1
291 2
```

6. Kinematics dictionary

Walking trials: **86** trajectories. Static trials: **63** (markers + angles only; no joint centers or COMs).

Units: marker/COM coordinates in **millimetres**; joint angles in **degrees** (Plug-in Gait: flexion/extension, abduction/adduction, rotation).

Markers (37)

Group	Names
Head	LFHD RFHD LBHD RBHD
Trunk	C7 T10 CLAV STRN RBAK
Left arm	LSHO LUPA LELB LFRM LWRA LWRB LFIN
Right arm	RSHO RUPA RELB RFRM RWRA RWRB RFIN
Pelvis	LASI RASI LPSI RPSI
Left leg/foot	LTHI LKNE LANK LHEE LTOE
Right leg/foot	RTHI RKNE RANK RHEE RTOE

Joint angles (26)

Group	Names
Lower limb	L/R Hip, Knee, Ankle, AbsAnkle; FootProgress
Pelvis / trunk	L/R Pelvis, Spine, Thorax
Head / neck	L/R Neck, Head
Upper limb	L/R Shoulder, Elbow, Wrist

Full angle field names: LHipAngles, LKneeAngles, LAbsAnkleAngle, LAnkleAngles, RHipAngles, RKneeAngles, RAnkleAngles, RAbsAnkleAngle, LPelvisAngles, RPelvisAngles, LFootProgressAngles, RFootProgressAngles, RNeckAngles, LNeckAngles, RSpineAngles, LSpineAngles, LShoulderAngles, LElbowAngles, LWristAngles, RShoulderAngles, RElbowAngles, RWristAngles, RThoraxAngles, LThoraxAngles, RHeadAngles, LHeadAngles.

Joint centers and COM (23) — walking trials only

Group	Names
Joints	LHJC RHJC LKJC RKJC LAJC RAJC
Whole body	CentreOfMass, CentreOfMassFloor
Segments	PelvisCOM, Left/Right Femur Tibia Foot, ThoraxCOM, HeadCOM, Left/Right Humerus Radius Hand

7. Precomputed gait metrics

Stored as MATLAB table objects. Readable in MATLAB; not numeric arrays in SciPy.

Trial Res

Field	Likely meaning
StpLen	Step length
StpWth	Step width
WkVel	Walking velocity
MTC	Minimum toe clearance
FCKneeAtt	Knee attitude at foot contact
MSKneeAtt	Knee attitude at mid-stance
MxStKneeAtt	Max stance knee attitude
Upright / nUpright	Trunk uprightness (raw / normalized)
NeckAng	Neck angle

Subject Res / RawRes extras

Field	Where
KneeAtt	Subject Res (collapsed)
Cadence	Subject Res and RawRes
Smooth / Smoothi	Subject Res (smoothness)
Coord	Subject Res (coordination)
FC/MS/MxStKneeAtt	RawRes (per trial)

8. Subject roster

Join labels to gait with S# (Subject No). Roster is the Excel No column.

Roster	S#	Vic	Type	Times	Cyber	Mass kg	Ht cm	WU trials	Frames	FC	Notes
1	S2	N	No	0	No	85.9	162.6	9	303–446	5–7	
2	S3	Y	Nd	2	Nd	47.7	162.6	8	280–379	5–7	WU11 instead of

Roster	S#	Vic	Type	Times Cyber	Mass kg	Ht cm	WU trials	Frames	FC	Notes
										WU08
	3 S4	Y	Both	4 No	53.6	162.6	9	220–291	4–5	WU10 not WU09
	4 S5	Y	Nd	2 No	52.7	167.6	8	214–270	3–5	no WU06
	5 S7	Y	Nd	3 Nd	59.5	161.3	8	217–318	3–7	no WU08
	6 S8	Y	Ip	3 No	85.5	163.8	9	253–330	4–6	
	7 S9	N	No	0 No	64.5	165.1	9	268–353	5–7	
	8 S11	N	No	0 No	70.9	161.3	9	204–290	4–5	
	9 S12	N	No	0 No	68.9	152.4	7	146–316	3–5	WUCopy + extra WKCopy; staticCopy
	10 S13	Y	Nd	4 Nd	100.0	165.1	5	262–320	5–5	fewest WU trials
	11 S14	Y	Both	12 Yes	54.5	160.0	8	209–383	5–7	no WU02; max times=12
	12 S15	Y	Both	10 Yes	64.1	141.0	9	302–384	6–8	shortest stature
	13 S17	N	No	0 No	60.5	157.5	9	300–375	6–7	WU0, WU3–WU7, WU9, WU11, WU13
	14 S19	Y	Ip	Nd Yes	73.6	162.6	4	168–224	2–4	fewest WU; no WU01
	15 S23	N	No	0 No	53.6	158.8	9	297–370	6–7	
	16 S26	Y	Nd	Nd Yes	71.4	162.6	9	293–460	5–8	longest trial 4.60 s
	17 S27	Y	Ip	2 Nd	51.8	157.5	10	253–349	4–7	
	18 S30	N	No	0 No	53.6	157.5	5	117–243	2–4	shortest trial 1.17 s
	19 S31	Y	Nd	1 Nd	65.9	167.6	9	131–415	2–7	gaps; WU10/11/13
	20 S32	Y	Ip	6 No	60.5	156.2	11	178–255	2–4	
	21 S34	Y	online	1 Yes	74.1	152.4	9	260–333	5–7	only online-only case
	22 S35	N	No	0 No	63.6	161.3	12	171–280	2–5	most WU trials

Roster	S#	Vic	Type	Times	Cyber	Mass kg	Ht cm	WU trials	Frames	FC	Notes
23	S37	Y	Ip	2	No	63.2	156.2	9	251–373	4–7	
24	S38	Y	Ip	5	Yes	66.8	160.0	9	246–343	5–7	
25	S39	N	No	0	No	51.4	162.6	10	281–445	5–8	
26	S40	N	No	0	No	61.4	165.1	8	257–393	4–7	no WU06/08/10
27	S41	N	No	0	No	52.3	154.9	8	201–314	4–6	
28	S42	N	No	0	No	72.3	162.6	6	377–421	6–8	
29	S43	N	No	0	No	68.6	161.3	6	236–290	4–6	
30	S46	Y	Nd	2	Nd	52.7	158.8	9	194–378	4–7	WU11, WU14
31	S48	N	No	0	No	63.2	160.0	10	236–313	4–6	no WU06/08

WU trial counts above are walking WU* trials only. S12 also has 10 extra WK*Copy takes that contain kinematics only (no event Info / Res).

9. MATLAB access

```
load('data/processed/Data_structure_all_subs.mat', 'Dat')

Dat.S14.Survey % Labels for subject 14
Dat.Survey % all 31 rows
Dat.S14.New_Session.WU01.kinematics.LKneeAngles % (frames × 3)
Dat.S14.New_Session.WU01.Info.KinFC % foot-contact events
```

10. Python access (SciPy)

Kinematics and event arrays load as NumPy.

Survey strings and all Res tables are MATLAB MCOS objects — not numeric.

For labels in Python, read data/raw/Victimization_surveys.xlsx and join on Subject No, or export Dat.Survey to CSV from MATLAB.

```
from scipy.io import loadmat
import pandas as pd

mat = loadmat('data/processed/Data_structure_all_subs.mat',
              struct_as_record=False, squeeze_me=True)
knee = mat['Dat'].S14.New_Session.WU01.kinematics.LKneeAngles # ndarray

survey = pd.read_excel('data/raw/Victimization_surveys.xlsx')
# join on Subject No == 14
```

11. Known irregularities

Item	Detail
S12 naming	Trials are WU01Copy... plus WK01Copy... (kinematics only). Calibration is staticCopy.
S17 naming	WU0, WU3–WU7, WU9, WU11, WU13 — not zero-padded WU01.
Missing WU numbers	Several subjects skip trials (S14 has no WU02; S19 starts at WU02).
VictimType coding	Free text: Ip / online / Both / Nd / No (online is lowercase).
Times	Sometimes Nd even when Victimized = Y (S19, S26).
MATLAB tables	Res, RawRes, Dat.Survey, posture refs, and Survey strings need MATLAB to read as data.
No analog forces	FPrate is stored; force-plate channels are not in this file.
S15 height	140.97 cm is an outlier vs the rest of the cohort (152–168 cm).

12. ML notes

- Natural binary target: Survey.Victimized (17 Y vs 14 N).
- Features: kinematics and/or event timing, unless Res tables are exported from MATLAB first.
- $n = 31$ is small. Split **by subject** (leave-one-subject-out or nested CV). Do not randomly split trials, or the same person will leak into train and test.